# Prove the fact, not your ID.

Zero-knowledge credentials from the IC chip in Japan's My Number Card, verified in any browser and on Solana.

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THE PROBLEM · WORLDWIDE

# ID documents are leaking everywhere.

US & CANADA · SEP 2026

**150M+**

records from an ID-verification vendor: driver's licences, passport numbers

JAPAN · SEP 2026

**1.6M**

ID images from a car-sharing service's sign-ups

GLOBAL · OCT 2025

**~70k**

government ID photos sent for age-check appeals on a chat platform

US · JUL 2025

**13k**

selfies and photo IDs uploaded to verify accounts on a dating-safety app

AUSTRALIA · MAR 2023

**7.9M**

driver's licence numbers (and 53k passport numbers) from a consumer lender

**Every one was a copy someone was asked to upload.**

Sources: TechCrunch (Sep 2026; 150M+ as reported, company confirmed licence and passport data), company disclosures and NHK (Japan), TechCrunch / NBC (Oct 2025), Reuters via Fortune (Jul 2025), Latitude Financial (Mar 2023).

THE PROBLEM · IT IS ABOUT TO GROW

# Laws now demand age and ID checks — and most checks still mean uploading your ID.

**UK · JUL 2025**

Online Safety Act: "highly effective" age checks for adult sites

**AUSTRALIA · DEC 2025**

Minimum age 16 for social media accounts

**US · 2026**

About half of the states mandate some form of age gating

**JAPAN · APR 2027**

Remote KYC moves from photo uploads to IC-chip reading

**More checks × stored copies**
**= more breaches.**

**$4.44M**

average cost of one data breach, worldwide (2025)

Sources: Ofcom; eSafety Commissioner; Jumio age-verification law snapshot (2026); JAFIC (NPA); IBM Cost of a Data Breach Report 2025.

CLOSE-UP · JAPAN, SEPTEMBER 2026

# A car-sharing service lost its members' ID cards.

**1.6M**

images of driver's licences and other ID documents uploaded at sign-up

**6.6M**

member accounts: name, address, date of birth, licence data

**30.6M**

people hit by leaks at Japanese listed companies in 2025 alone

Sources: company disclosures (Sep 2026), NHK (Oct 2026), Tokyo Shoko Research (Jan 2026). Police are reissuing affected licences.

# Data you never collect cannot leak.

TODAY

**`collect → store → leak`**

Every service keeps a copy of your ID.

WITH P!PON

**`verify → forget`**

The service checks a proof and keeps nothing.

This is the turning point: services will stop holding personal data wherever they can.

WHY START IN JAPAN

# Japan is moving identity checks from photos to the chip.

## Apr 2027

Remote KYC under the anti–money-laundering law drops photo uploads in favour of reading the IC chip.

## 104.5M

My Number Cards held (84.5% of the population). Each chip can carry a certificate signed by the national authority, J-LIS.

Sources: National Police Agency (JAFIC) Q&A on the amended ordinance; Ministry of Internal Affairs and Communications, cards held at end of Sep 2026.

# IC chips solve authenticity.
# Zero-knowledge proofs solve privacy.

**My Number Card**

Certificate signed by J-LIS. Signs a fresh challenge.



**Phone**

Checks the signatures inside a zero-knowledge proof.



**Website or Solana**

Verifies the proof. Gets a yes.

**The service learns**

Over 20 · lives in a prefecture · hasn't claimed before

**It never sees**

Name · date of birth · address · ID image · My Number

WHO IT IS FOR

# Every service that asks for an ID only to learn one fact.

**Online stores**
Alcohol, tobacco, adult content

**Sign-up checks**
"A real, unique person"

**One per person**
Drops, airdrops, giveaways

**DAO votes**
Sybil-resistant governance

**AI agents**
Acting for a verified human

**Not legal KYC**
It's for everything else

WHY SOLANA

# A shared "one per person" registry that no company has to own.

**~144k** compute units: Groth16 verify (alt_bn128), nullifier and NFT mint in **one transaction**

### One per person, no list of people

Each card leaves a per-gate nullifier in a PDA. A second claim fails on-chain.

### A trust list anyone can read

Gates hold the trusted J-LIS CA keys on-chain. Our Web2 store reads the same list: no API key, no vendor.

### Next: attestations and agents

Solana Attestation Service for reusable credentials; x402 so an AI agent can show a verified human stands behind it.

Live on devnet: zk_gate program with real-card and demo gates · see the technical demo

WHY NOT THE FREE TOOLS?

# Passport-based ZK tools miss most of Japan.

| | Passport-based tools (Self, ZKPassport) | p!pon |
|---|---|---|
| **ID people hold** | Passport: 17.8% of Japanese | **My Number Card: 84.5%** |
| **Where you live** | No address on the passport | **Prefecture now, city next** |
| **Japanese rules** | Global, generic | **Built for the 2027 IC-chip KYC shift; path to certified revocation** |
| **Buying in Japan** | Open source, self-serve | **Japanese support, SDK, legal documents** |
| **Abroad** | 127+ countries' passports | **Same standard next (ICAO 9303), plus a shared Solana registry** |

Sources: MOFA passport statistics (end of 2025); MIC card statistics (Sep 2026); Self blog (Sep 2025: passports of 127 nations, ID cards of 35 countries, Aadhaar; Japan's card not listed); ZKPassport listing.

ROADMAP

# From one card in Japan to every chip, for people and their agents.

NOW · OCT 2026

## Working sample

My Number Card: age + prefecture + one per person

zk_gate on Solana devnet; browser verifier for Web2

iPhone on-device proof (9 s); Mac + USB reader

Tested end to end on a real card

NEXT · BY Q1 2027

## Product for stores

iPhone app in TestFlight, then Android

Drop-in checkout SDK for online stores

More facts: age lines, city, recent certificate

Solana Attestation Service; multi-party setup, audit, mainnet

AI AGENTS · 2027

## A human behind the agent

Delegate a scoped, revocable "verified human, 20+" credential to your agent

Proof attached to x402 payments for age-restricted goods

Per-human limits via nullifiers: one person, N agents

Tools for agent wallets and MCP

GLOBAL · 2027+

## Every ID chip

ePassports to the ICAO 9303 standard: chip data signed by each country's CA

Other national eID cards with signed chips

Same proof, same Solana registry, any country

Revocation through certified operators

ROADMAP · PRODUCT DESIGN · WEB3

# Scan once. After that, just sign with your wallet.

ONCE · ABOUT A MINUTE

**1.** Tap the card; the proof "20+, one per person" is bound to your wallet.

**2.** zk_gate verifies it on-chain and issues a **non-transferable attestation** to the wallet (Solana Attestation Service), with an expiry.

EVERY TIME AFTER · SECONDS

**3.** A service asks the wallet to sign — no card, no password.

**4.** It reads the attestation on-chain: issued by zk_gate, not expired, held by this wallet.

### Bound to the wallet

Non-transferable: it can't be sold or lent.

### It expires

Re-scan yearly or when the certificate renews; address facts sooner.

### One per person

The nullifier stops a second wallet for the same card.

### Your choice of linkage

One wallet everywhere is convenient; separate wallets keep services apart.

Planned. Today's demo proves on every request; the attestation turns that one proof into a reusable credential.

ROADMAP · PRODUCT DESIGN · WEB2

# Verify at sign-up. After that, just log in.

**ONCE · AT SIGN-UP**

**1.** Tap the card; the proof is bound to the store's sign-up session.

**2.** The store verifies it and saves on the account only: **"20+ verified on (date)"** and its store-only pseudonymous ID.

**EVERY TIME AFTER · USUAL LOGIN**

**3.** Log in with a password or passkey, as today.

**4.** Checkout reads the flag on the account: alcohol can be sold, no ID asked.

### Nothing to leak

A breach exposes a yes/no flag, not a licence image.

### One account per person

The pseudonymous ID blocks duplicate sign-ups.

### Re-check when it matters

Yearly, or before a high-risk order, by the store's policy.

### Or reuse across stores

A store can also accept the Web3 attestation: no scan at all.

Works with today's verifier: the store keeps the result of one proof instead of a copy of the ID.

ECOSYSTEM AND POLICY

# Make ZK proofs lawful in Japan — as one industry.

THE COMPANY

**p!pon**

Builds the product, the SDK and the business. One member of the association — not its owner.

AN INDUSTRY ASSOCIATION · CO-FOUNDED WITH PEERS

**ZK identity projects (any chain), accelerators, consultancies**

Study sessions and guidelines · briefings for regulators (AML, Digital Agency, J-LIS, privacy) · joint legal clarifications · shared trust lists and specs

## Good for everyone in Japan

Once a ZK proof is a lawful way to check age or residence, no service needs your ID copy. Fewer copies, fewer leaks — for all 124 million people, not just our users.

## A precedent for the world

A national PKI on 100M+ cards and IC-chip KYC from 2027: a legal status for ZK proofs in Japan gives other countries a model — and speeds up blockchain adoption there.

Plan: study group from late 2026; association in early 2027; first output: a data-minimising age-check guideline for the April 2027 IC-chip shift.

STATUS

## Working today on devnet, end to end with a real My Number Card.

Next: stores, AI agents, then every ID chip — see the roadmap.

# Stop collecting IDs.
# Prove the fact.



TECHNICAL DEMO · ARCHITECTURE

# No issuer. The national CA is the root of trust; the phone does the proving.

| Layer | Built with |
|---|---|
| Circuit | Circom · Groth16 · 1.49M constraints |
| Prover | mopro + rapidsnark on iPhone · Rust CLI on Mac |
| Card I/O | Core NFC (iPhone) · CryptoTokenKit / PC/SC (Mac) |
| On-chain | Anchor program zk_gate · alt_bn128 · Metaplex Core |
| Web verifier | snarkjs in the browser · CA trust list read from Solana |

INSIDE THE PROOF

# Four checks in private. This sample reveals two facts.

**1.** J-LIS CA signature over the certificate (RSA-2048, SHA-256)

**2.** The card's signature over a challenge bound to the wallet or browser session

**3.** Date of birth + N years ≤ today

**4.** Address starts with the chosen prefecture

**Revealed in this sample**

**Age ≥ 20 · Prefecture**

Plus the plumbing: CA key hash, a per-service nullifier, challenge, date, scope.

**Room to grow — the certificate also holds name, date of birth, sex and full address, so the same pattern can prove (not built yet):**

- Other age lines: 18+, 65+, under 25
- City or ward
- Sex, only where a service must know
- Name matches the account (hash, not the name)
- Certificate issued within N days

FLOW 1 · WEB2 · ONLINE STORE SELLING ALCOHOL

# How the store learns "20+" without seeing your ID

STORE

### 1 · Checkout asks

Shopper taps "Verify age with p!pon". The store makes a one-time session ID and shows it as a QR code (a link on mobile).

PHONE

### 2 · Request opens

The p!pon app shows who asks and what: "Kura Online Store · Age 20+".

CARD

### 3 · Tap the card

After the signing password, the chip signs SHA-256(session ID ‖ nonce) and gives its J-LIS-signed certificate.

PHONE

### 4 · Prove on device

Groth16 proof (~9 s): CA signature, card signature, age ≥ 20, prefecture. The certificate never leaves the phone.

STORE'S BROWSER

### 5 · Verify, ~200 ms

Proof valid · same session (no replay) · CA in the trust list on Solana · dated today · 20+ · scoped to this store.

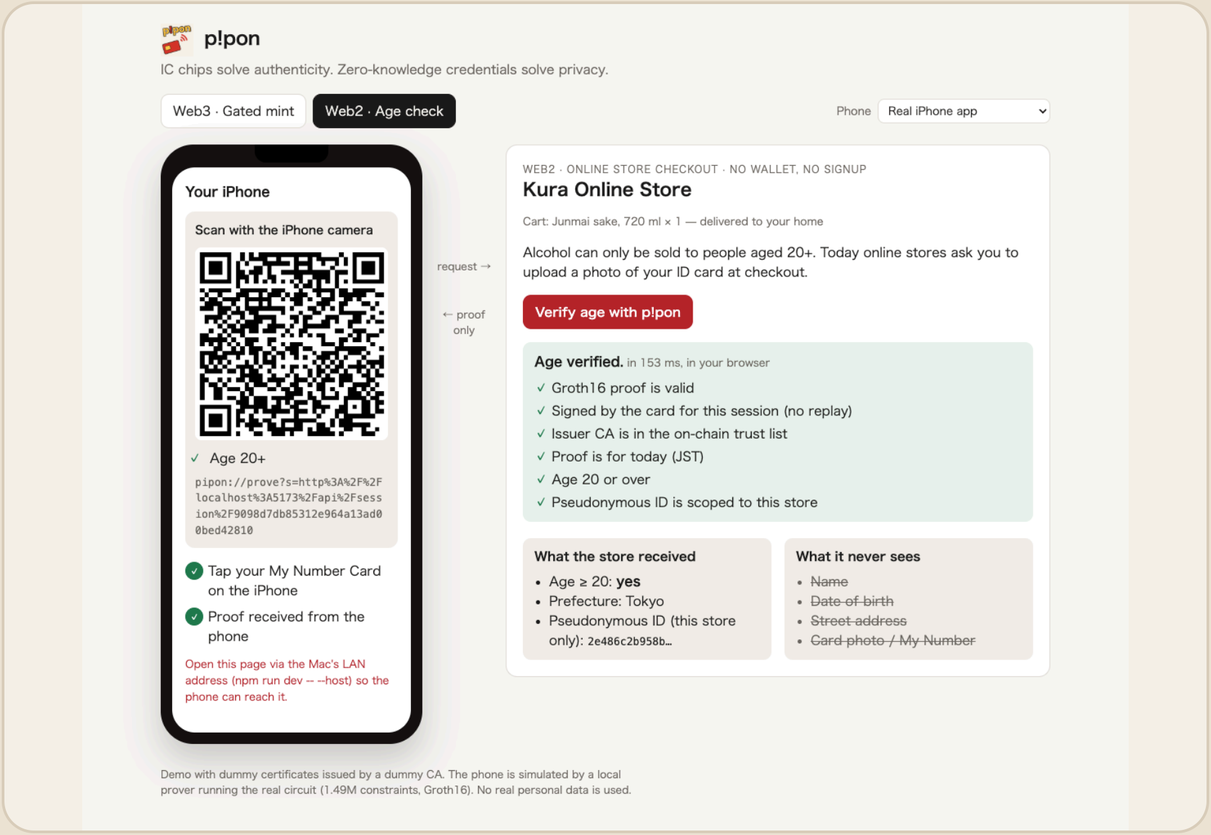
**Store keeps:** age 20+ = yes, prefecture, a store-only pseudonymous ID.

**Never receives:** name, date of birth, address, ID image, My Number.

Once per account or order, from home — not a card scan at a shop counter.

DEMO 1 · WEB2 · ONLINE STORE CHECKOUT

# No wallet, no signup: the store gets a yes.




An online store selling sake: age 20+ needed at checkout

Verified in the browser in ~200 ms; issuer CA checked against the trust list on Solana

Store learns: 20+, prefecture, a pseudonymous ID for this store only

FLOW 2 · WEB3 · ONE-PER-PERSON MINT

# How Solana enforces "20+, one per person"

WALLET

### 1 · Open the gate

The gate account on Solana holds the rules: minimum age, prefectures, trusted CA keys, scope.

PHONE

### 2 · Bound to the wallet

The challenge is SHA-256(wallet address ‖ nonce), so the proof only works for this wallet.

CARD + PHONE

### 3 · Tap and prove

The card signs the challenge; the phone makes the same Groth16 proof as in Web2, with the gate's scope.

SOLANA · zk_gate

### 4 · Verify on-chain

Groth16 via alt_bn128 syscalls; CA, age, prefecture, date and the signer's wallet checked against the gate.

SOLANA · zk_gate

### 5 · Mint or reject

New nullifier → PDA created, Core NFT minted, same transaction (~144k CU). Nullifier exists → "Already claimed".

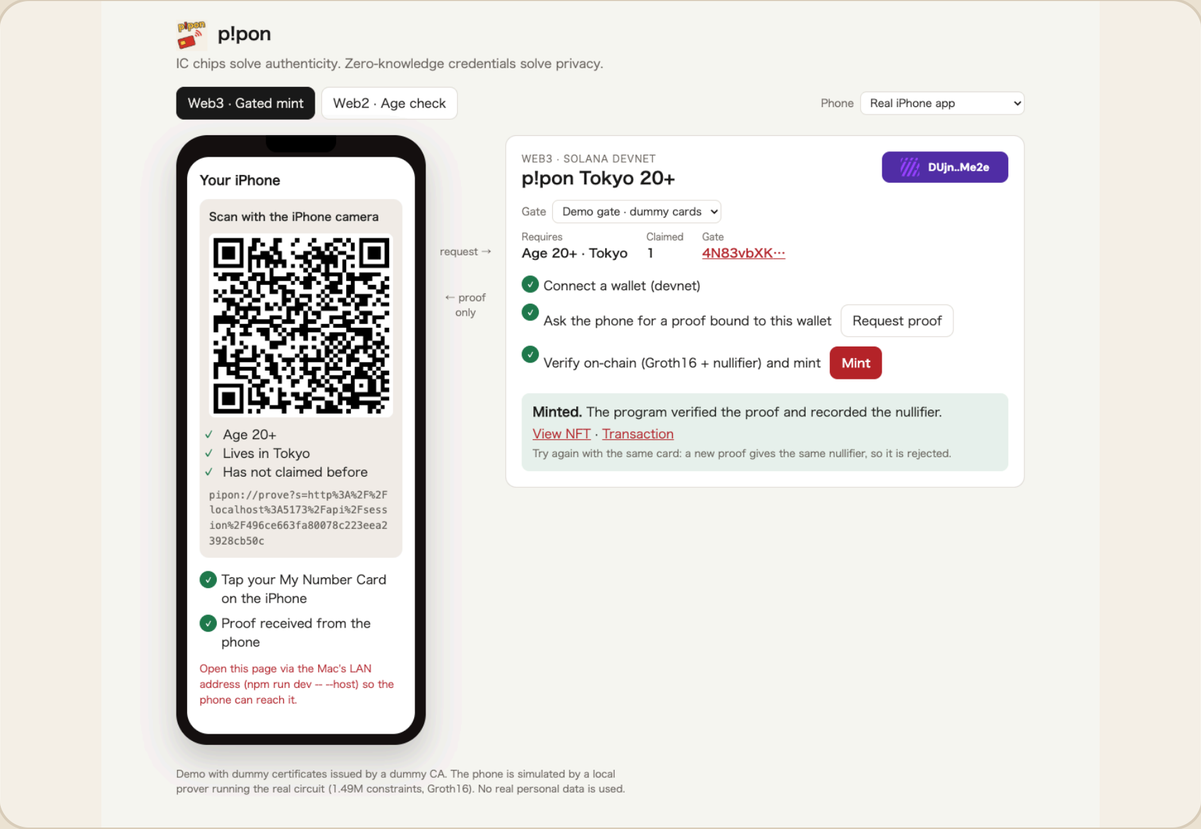
**Public on-chain:** the proof and its values — CA key hash, nullifier, date, age threshold, prefecture code.

**Never on-chain:** name, date of birth, address, ID image, My Number.

The nullifier is per card and per gate: another gate gets a different one, so wallets can't be linked across services.

DEMO 2 · WEB3 GATED MINT

# One transaction: verify, record, mint.




Proof bound to the wallet

Same card again → **Already claimed**

Card from another prefecture → **rejected by the gate**

TESTED ON A REAL MY NUMBER CARD

# Same flow, real J-LIS certificate.

**3.5 s**

proof on a Mac mini, card in a USB reader

**9 s**

on-device proof on an iPhone 15 Pro Max

**2 / 2**

claim verified on devnet · second claim from a new wallet rejected

Devnet test gate (scope 3), 2026-10-07 · tx c623tn…zvYv7 · no personal data shown or stored

LIMITS AND NEXT

# Honest about what's left.

## Limits today

No revocation check yet: revocation lists are only for certified verifiers.

Single-contributor trusted setup: demo only.

## How the roadmap closes them

Revocation through a certified platform operator.

Multi-party trusted setup and an audit before mainnet.

iPhone NFC in TestFlight (paid signing team pending).